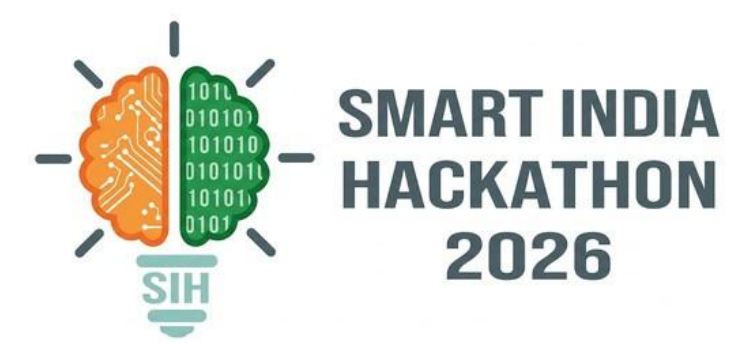


SMART INDIA HACKATHON 2026

JAL DRISHTI



Sense the Storm. See the Risk. Act in Time.

Problem Statement ID: SIH26192

Problem Statement Title: Multi-Source Flash Flood Prediction for Hilly Regions

Theme: Disaster Management

Category: Software

Team ID: - 159135

Team Name: The Singularity



THE PROBLEM

FLOOD SIGNALS ARE FRAGMENTED

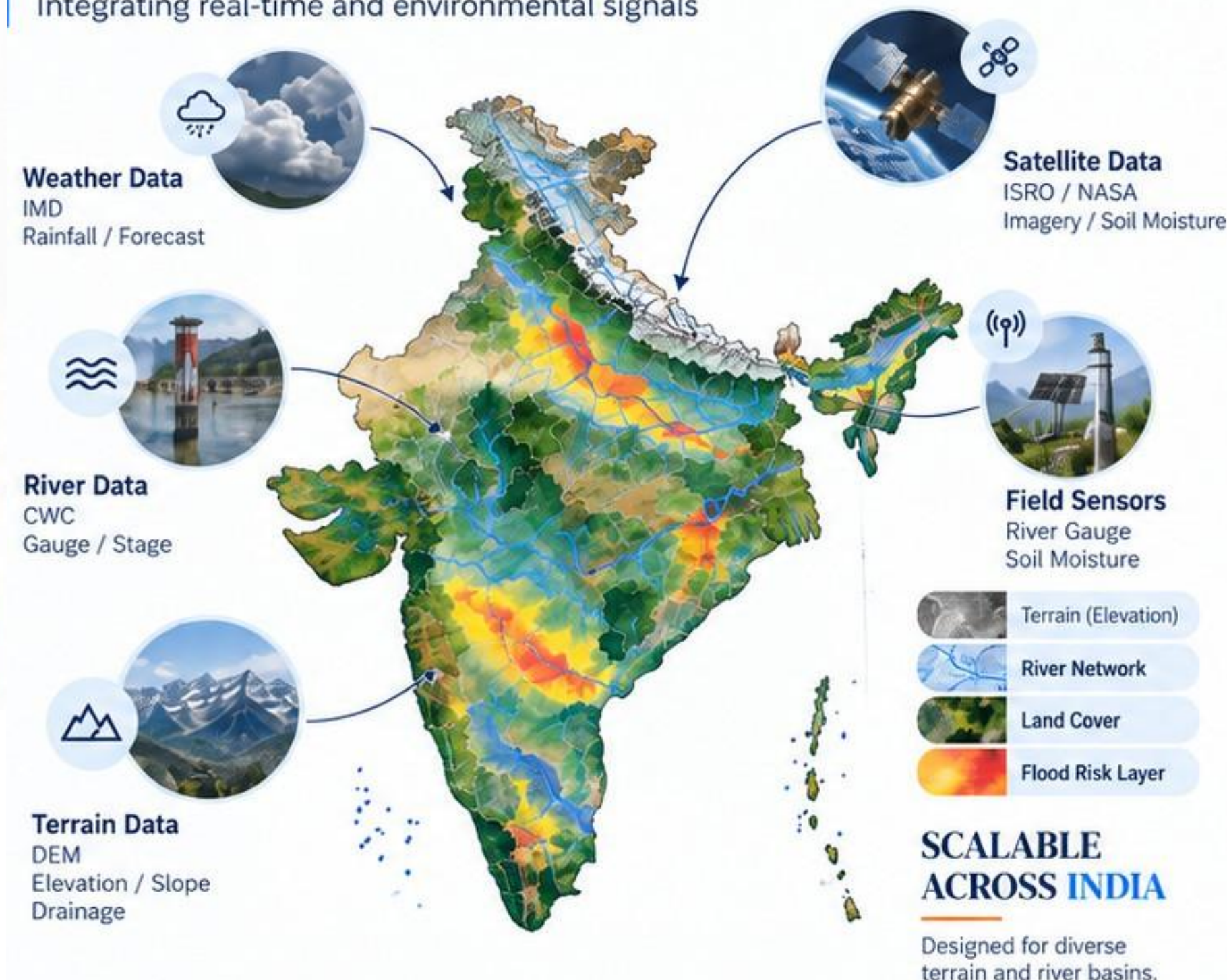
In hilly regions, flood risk changes rapidly due to intense rainfall, river conditions, terrain and environmental factors. These signals are available from multiple sources but are **analysed separately**, making it difficult to generate timely, **location-specific** decisions for early warning and response.



Jal Drishti unifies these signals through ML, hydrological modelling and geospatial analysis to generate location-specific flood-risk intelligence.

MULTI-SOURCE INPUTS

Integrating real-time and environmental signals



DATA INPUT + HYDROLOGICAL ANALYSIS → LOCALIZED RISK → DECISION SUPPORT

TECHNOLOGY STACK

Frontend

- React 18.3.1, Vite 5.4.2
- Tailwind CSS 3.4.10
- Leaflet 1.9.4, React-Leaflet 4.2.1
- Three.js / React Three Fiber
- Recharts

Backend

- Python 3.11, FastAPI 0.115
- Uvicorn
- Pydantic v2
- Pandas, NumPy, PyArrow

ML & Hydrology

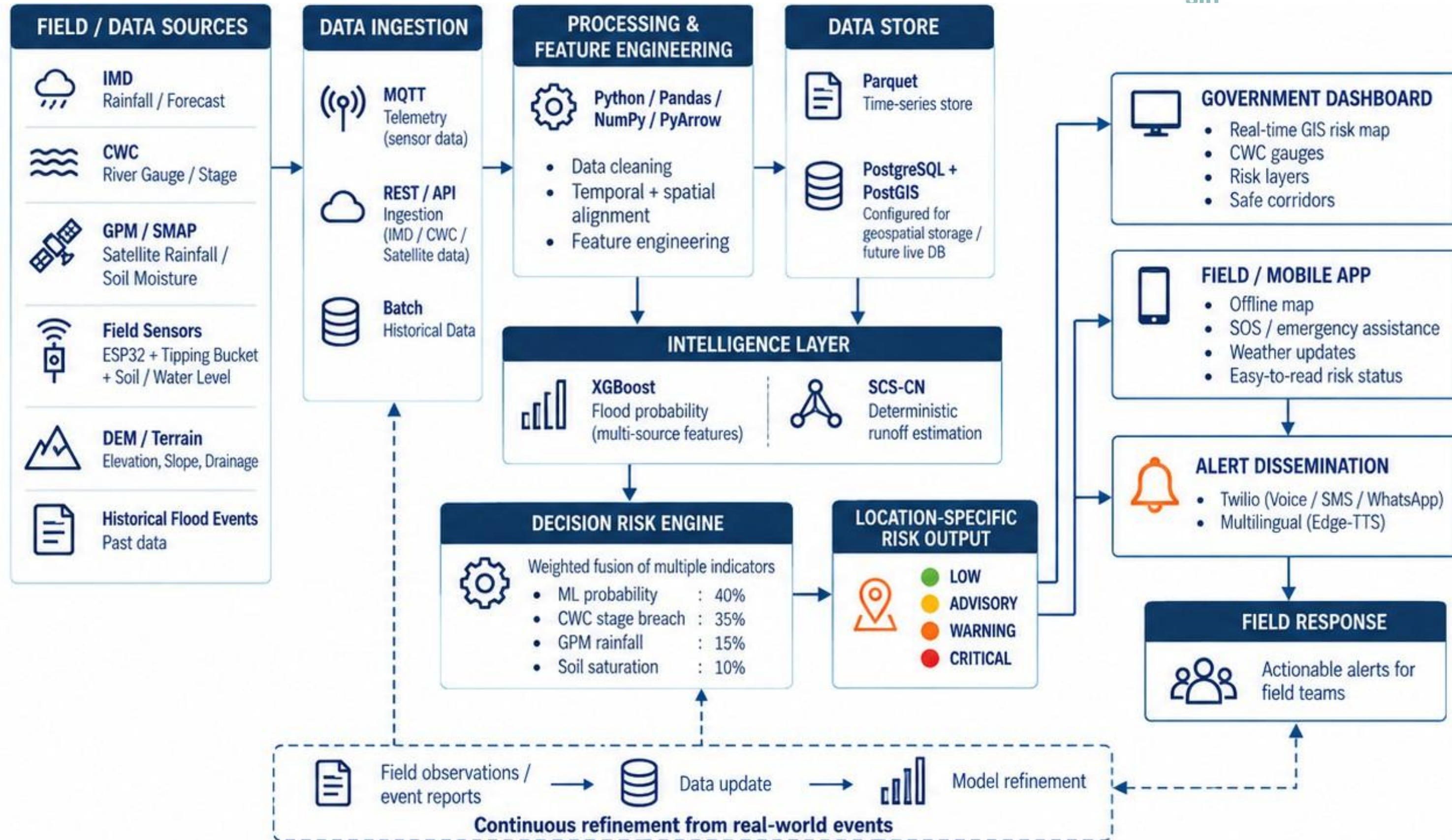
- XGBoost 2.1.0
- Scikit-Learn
- SCS-CN
- PyTorch benchmarks

Data & Storage

- Parquet
- PostgreSQL + PostGIS (configured/partial)

IoT & Alerts

- ESP32-WROOM-32
- Tipping Bucket Rain Gauge
- JSN-SR04T
- MQTT
- Twilio
- Edge-TTS



01 | FEASIBILITY

Why it can be built and deployed

- Uses existing and open data sources (IMD, CWC, ISRO/NASA, DEM)
- Leverages low-cost, proven sensors (ESP32 + rain gauge + soil moisture)
- Built with open-source technologies (MapLibre GL, PostgreSQL, FastAPI, Python)
- Modular architecture for phased deployment
- Works in low-connectivity environments (offline mesh, IVR)
- Can be piloted in high-risk districts and scaled to other regions

02 | VIABILITY

Why it is sustainable and impactful

- Addresses a real and recurring problem (increasing flash flood events in Uttarakhand)
- Supports government decision-making with actionable, location-specific insights
- Cost-effective and scalable using open-source stack and existing infrastructure
- Adaptable to multiple regions and hazards (landslides, heavy rainfall)
- Enables long-term disaster resilience through continuous data integration and learning
- Aligns with national initiatives (NDMA, CWC, SDMA, Digital India)

KEY CHALLENGES

- Limited real-time data from remote areas
- Fragmented data across multiple agencies
- Difficult terrain and poor connectivity
- Delayed decision-making
- Lack of localized, actionable information

OUR SOLUTION

- Low-cost sensors + offline mesh network
• Real-time data collection & transmission
- Unified data pipeline
• Geospatial database integration
- Offline maps, IVR alerts
• Lightweight deployment (optimized for low bandwidth)
- Real-time risk analysis and visualization
• Interactive dashboard & alerts
- District-level risk maps
• Prioritized and location-specific alerts

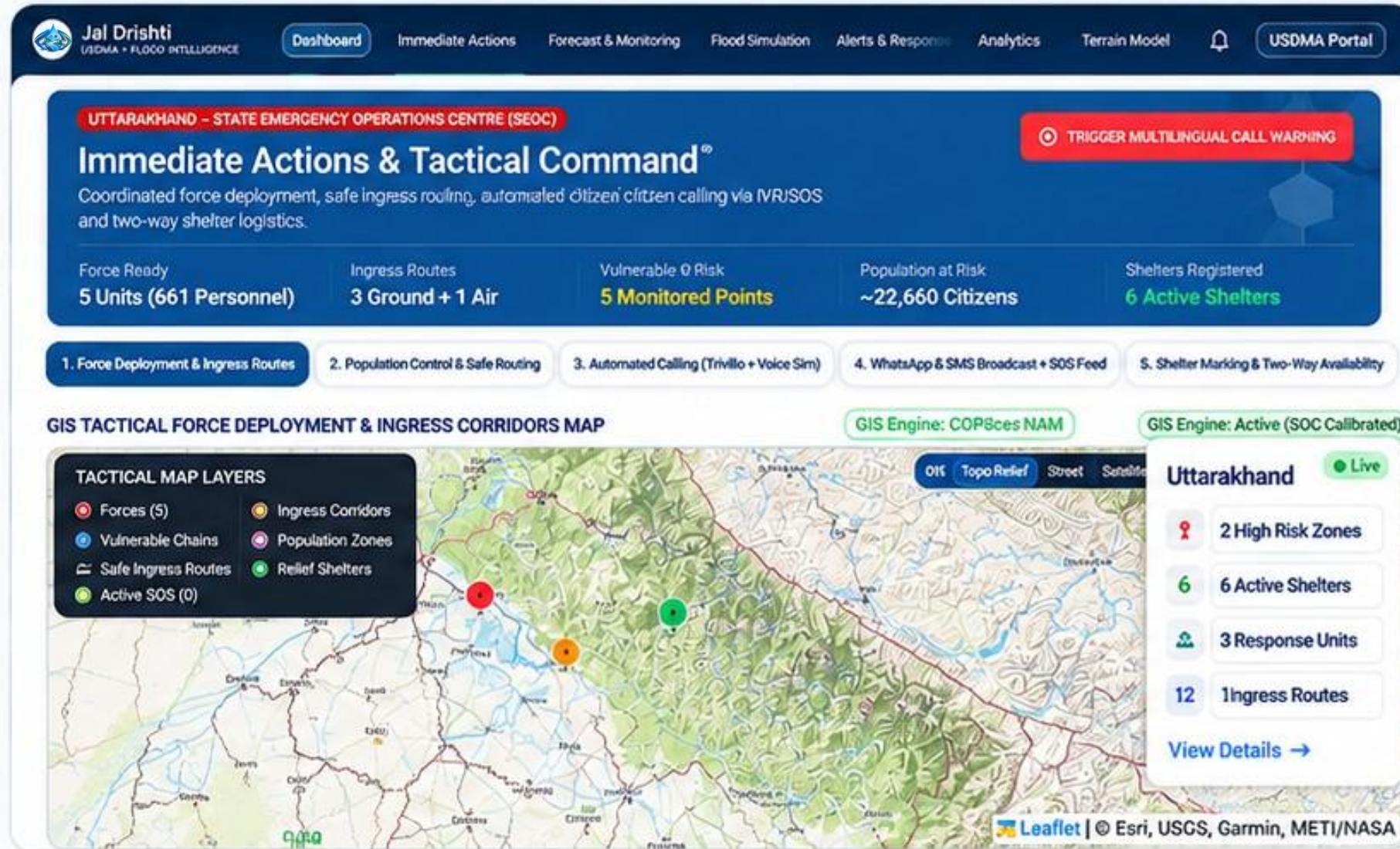


IMPACT AND BENEFITS

Safer Communities. Faster Response. Smarter Decisions

1 WEB PLATFORM

Real-time monitoring, risk intelligence and coordinated response



4 KEY BENEFITS

Tangible value for all stakeholders



For Government

- Data-driven decision making
- Efficient resource deployment
- Real-time situational awareness



For Communities

- Timely warnings and safer lives
- Access to local information (offline)
- Inclusion through multilingual support



For Field Teams

- Accurate risk maps and safe routes
- Offline tools for low-connectivity areas
- Faster and better-coordinated response

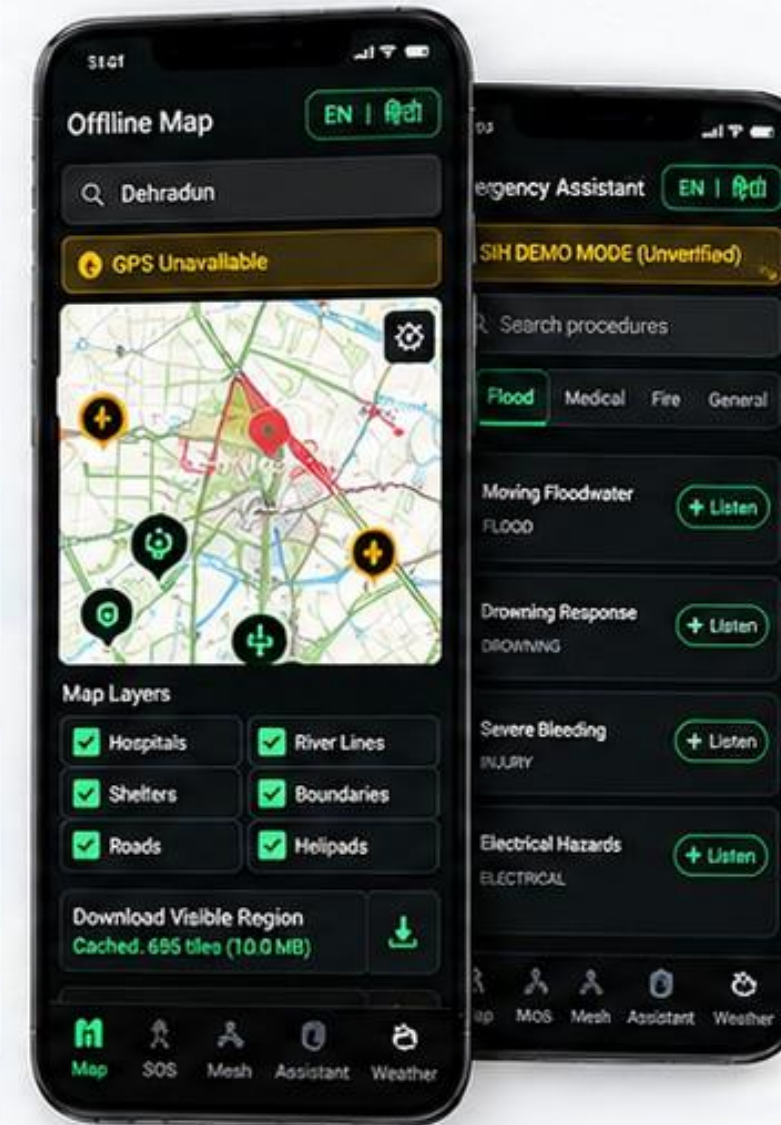


For Environment & Infrastructure

- Reduced loss of life and property
- Better floodplain management
- Long-term resilience through data and models

2 FIELD APP (MOBILE)

Offline maps, SOS, emergency assistant and local language support



3 MEASURABLE IMPACT

Based on prototype simulations and pilot scenarios

Before With Jal Drishti

Early Warning Lead Time

More time for preparation and evacuation



Response Mobilization Time

Faster deployment to affected areas



At-Risk Population Coverage

More people reached through targeted alerts



01 KEY RESEARCH PAPERS

1. Saharia et al. (2021). India flood inventory: creation of a multi-source national geospatial database to facilitate comprehensive flood research.
<https://doi.org/10.1007/s11069-021-04698-6>
2. Dimri et al. (2017). Cloudbursts in Indian Himalayas: A review.
<https://doi.org/10.1016/j.earscirev.2017.03.006>
3. Kumar et al. (2018). Hydrometeorological aspects for cloudburst and flash flood events in the Third Pole region.
<https://doi.org/10.1016/j.polar.2018.08.004>
4. Hasanuzzaman et al. (2023). Characterizing recurrent flood hazards in the Himalayan foothill region.
<https://doi.org/10.1016/j.asr.2023.02.028>
5. Arora et al. (2025). Machine learning model optimization for flood susceptibility zonation over the Kosi megafan.
<https://doi.org/10.1038/s41598-025-07403-w>

02 DATASETS & OFFICIAL SOURCES

-  **India Flood Inventory (IFI-Impacts)**
HydroSense Lab, IIT Delhi (1967–2023)
[Zenodo / GitHub](#)
-  **Bhuvan Disaster Services**
NRSC / ISRO – hazard mapping & monitoring
[Bhuvan](#)
-  **India Meteorological Department (IMD)**
Rainfall & official flood guidance
mausam.imd.gov
-  **CivicDataLab – Himachal Pradesh**
Open-source geospatial data & risk scoring
[GitHub](#)

03 COMPARISON WITH EXISTING SYSTEMS

Feature / Capability	Jal Drishti (Our Solution)	CWC / IMD (Official Systems)	ISRO Bhuvan	Open Source Tools
Integrated multi-source flood risk workflow	✓	—	—	—
ML-based flood probability model	✓	—	—	—
Geospatial visualization (2D/3D terrain)	✓	Limited*	✓	✓
Real-time monitoring & forecasting	In development (with sensors)	✓	Limited**	—
Offline field app (maps, SOS, resources)	✓	—	—	—
Field sensor telemetry (soil moisture, etc.)	✓	—	—	—
BLE mesh communication	✓	—	—	—
Multilingual emergency assistant	✓	—	—	—
Targeted alerts & tactical command dashboard	✓	—	—	—

04 LIVE DEMO & REPOSITORY



Live Demo
(Sample Interface)

https://drive.google.com/drive/folders/1r3Fdj-1gi80tSN_aeKDdhjy_x7ATRZR6C



Source Code
(GitHub Repository)

<https://github.com/saswata12223/JAL-DRISHTI.git>



05 SYSTEM INTERFACES



06 RESEARCH TO SYSTEM WORKFLOW

